Height-Aware Attenuation Priors for Dense UAV Channel Knowledge Maps

Guillem Moreno Garcia, Evgenii Vinogradov, Sergi Abadal
Universitat Politècnica de Catalunya (UPC), Barcelona, Spain
guillem.moreno.garcia@estudiantat.upc.edu

***Abstract*—Dense channel knowledge maps can replace repeated ray tracing in radio planning, but a useful surrogate must preserve the strong height, range, and visibility structure of air-to-ground attenuation. This paper presents a height-aware, auditable attenuation prior for dense unmanned aerial vehicle channel knowledge maps. A line-of-sight branch combines free-space loss with a coherent image-source two-ray term and training-only height/range calibration. A non-line-of-sight branch combines a COST231-shaped urban term with local morphology descriptors, followed by regime-specific linear calibration. The dataset-supplied geometric visibility mask selects the branch; it is the deterministic `LineOfSight` output of the configured ray tracer. The evaluation uses a strict city holdout with 2590 maps from 14 unseen cities, each containing 513×513 receiver locations at 7.125 GHz. The prior reaches 1.9277 dB overall root mean square error, with 1.7370 dB in line of sight and 3.5275 dB in non-line of sight, while attaining a per-map spatial correlation of 0.9481. A city bootstrap gives a 95% interval of 1.7960 dB to 2.0967 dB for overall error. The result provides a strong, inspectable starting point for fast UAV radio-map generation and for later residual learning.**

***Index Terms*—air-to-ground channel, channel attenuation, channel knowledge map, path loss, radio map, unmanned aerial vehicle.**

I. Introduction

Channel knowledge maps (CKMs) organize location-dependent channel information so that planning and adaptation do not require a new measurement or ray-tracing query for every link [1], [2]. The idea is especially relevant to unmanned aerial vehicle (UAV) networks, where transmitter height changes visibility, propagation distance, and the interaction with the urban skyline [3], [4]. A dense CKM surrogate must therefore reproduce more than a plausible image: it must preserve radial attenuation structure, line-of-sight (LoS) and non-line-of-sight (NLoS) regimes, and their dependence on height.

Data-driven radio-map methods can achieve accurate dense predictions [5], but an unconstrained regressor may spend capacity recovering simple propagation structure and may hide a weak NLoS regime behind a global metric. Purely analytical models present the opposite problem. Free-space and two-ray models capture range and coherent reflection, while classical urban models describe broad blocked-link trends [6]. Their original coefficients and deployment domains, however, do not directly reproduce the high-resolution ray-traced CKM maps considered here.

The specific gap addressed here is an inspectable attenuation anchor that combines these two views under a strict unseen-city protocol. The proposed prior keeps literature-derived propagation structure, but fits all CKM-specific coefficients using training cities only. LoS pixels use a coherent two-ray model with smooth height and radial calibration. NLoS pixels use a clipped COST231-shaped term plus local morphology features. A geometric visibility mask combines both maps.

The main contributions are:

- a height-aware LoS prior that preserves coherent radial rings instead of reducing the link to distance-only loss;
- a morphology-aware NLoS prior whose raw propagation estimate, local building descriptors, and fitted regime remain separately inspectable; and
- a leakage-resistant evaluation on 14 unseen cities, including an analytical component ablation, LoS/NLoS and height diagnostics, and uncertainty from a held-out-city bootstrap.

The scope is deliberately limited to the attenuation prior. Delay spread, angular spread, and neural residual learning are outside the contribution.

A. Related Work and Positioning

CKMs describe location-indexed channel side information rather than a single link budget. Their uses include environment-aware scheduling, beam selection, and trajectory or deployment planning [1], [2]. Dense attenuation-map prediction is commonly approached as image-to-image regression. RadioUNet, PMNet, and related challenge systems demonstrate the strength of convolutional encoder–decoder models when topology and sparse or simulated radio information are available [5], [7]–[9]. These methods motivate dense output, but many reported settings do not combine a continuously varying elevated transmitter with a strict city holdout.

Hybrid models address the complementary weakness of pure image regression: known propagation structure should not have to be rediscovered independently in every city. Free-space and coherent two-ray models explain the dominant range, height, and reflection structure of unobstructed links [10]–[12]. COST231/Hata models provide a coarse blocked-link trend [6]. However, their nominal coefficients and validity ranges cannot simply be transferred to a dense 7.125 GHz ray-traced dataset. We therefore retain their functional structure as a prior and fit only compact calibration terms on training cities.

This positioning differs from learning a correction to a single path-loss formula. The LoS branch explicitly preserves coherent rings, whereas the NLoS branch combines a conservative analytical scale with observed local morphology. It also differs from probabilistic LoS models: a deterministic visibility mask is available here and is used to route each receiver through the appropriate branch. The result is a fully dense attenuation estimate whose intermediate quantities remain accessible for diagnosis.

II. Data and Evaluation Contract

Each sample is a 513×513 urban CKM with $1\,\mathrm{m}$ spatial resolution. One elevated transmitter is horizontally centered in the map, operates at $7.125\,\mathrm{GHz}$, and has a sample-dependent height $h_{\mathrm{tx}} \approx 12\,\mathrm{m}$ to $478\,\mathrm{m}$. Every non-building pixel is a receiver candidate at $h_{\mathrm{rx}} = 1.5\,\mathrm{m}$. The metric first excludes all building pixels and then retains only finite ray-traced targets in the stored $[20, 185]$ dB support. The prediction target is the stored dense channel-attenuation map as an unsigned 8-bit array with 1 dB quantization.

Headline metrics use the binary `LineOfSight` field emitted by the configured MATLAB ray tracer and stored as `los_mask`. It is keyed by the city geometry, transmitter position and height, receiver grid, and ray-tracing settings, and is not inferred by the attenuation prior. Building pixels are removed before this mask defines LoS and NLoS receiver sets.

The split is made by city rather than by pixel, tile, or random sample. The official split contains 10840 maps from 37 training cities, 2750 maps from 15 validation cities, and 2590 maps from 14 final test cities. No city is moved between these partitions. All lookup tables, standardization statistics, and regression coefficients use training cities only. Validation is used for development checks; headline results and uncertainty intervals use the final test partition after every fitted quantity has been frozen.

For valid-receiver mask $m_i(p)$, prediction $\hat{y}_i(p)$, and target $y_i(p)$, the pixel-weighted test error is

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i \sum_p m_i(p)[\hat{y}_i(p) - y_i(p)]^2}{\sum_i \sum_p m_i(p)}}. \tag{1}$$

We also report mean absolute error (MAE) and map correlation. Let $\Omega_i = \{p : m_i(p) = 1\}$, $N_i = |\Omega_i|$, and let $\widetilde{\mathbf{y}}_i$ and $\widetilde{\hat{\mathbf{y}}}_i$ denote the valid target and prediction vectors after subtracting their respective per-map means. Then

$$\begin{aligned} \rho_i &= \frac{\widetilde{\hat{\mathbf{y}}}_i^{\top} \widetilde{\mathbf{y}}_i}{\|\widetilde{\hat{\mathbf{y}}}_i\|_2 \|\widetilde{\mathbf{y}}_i\|_2}, \\ \mathrm{MapCorr} &= \frac{\sum_{i \in \mathcal{C}} N_i \rho_i}{\sum_{i \in \mathcal{C}} N_i}. \end{aligned} \tag{2}$$

The set $\mathcal{C}$ excludes maps with fewer than two valid pixels or zero variance in either vector. Thus, map correlation is the valid-pixel-weighted mean of per-map Pearson correlations, not one correlation after concatenating all cities. RMSE remains the primary unit-aware metric; map correlation checks whether relative high- and low-attenuation regions are placed correctly. Uncertainty is estimated by 20000 percentile-bootstrap draws over the 14 final test cities. Each draw samples cities with replacement and recomputes the pixel-weighted metric; this preserves within-city spatial dependence rather than treating hundreds of millions of pixels as independent observations.

TABLE I
Evaluation setting.

Item	Setting
Map	513×513, 1 m pixels
Carrier	7.125 GHz
Transmitter	Centered; variable height
Receiver	Ground, 1.5 m; buildings masked
Visibility	Configured-ray-tracer LoS output stored in HDF5
Split	10840/2750/2590 train/validation/test maps by city
Test cities	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver

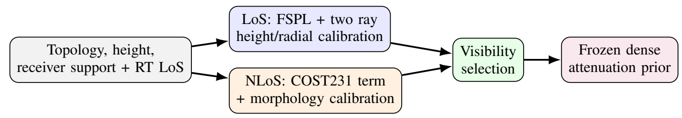

Fig. 1. The prior keeps LoS and NLoS propagation mechanisms separate. All CKM-specific calibration is fitted on training cities and then frozen.

III. Height-Aware Attenuation Prior

Fig. 1 summarizes calibration and deployment. The method uses known geometry to compute distances, a building/visibility mask to route LoS and NLoS pixels, and local windows over the building-height map. Only compact calibration tables and linear weights are learned. They are frozen before a held-out city is evaluated.

A. LoS Branch

Let d_{2D} be the horizontal transmitter-receiver distance. The direct and image-source reflected distances are

$$\begin{aligned} d_{\mathrm{LoS}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} - h_{\mathrm{rx}})^2}, \\ d_{\mathrm{ref}} &= \sqrt{d_{2D}^2 + (h_{\mathrm{tx}} + h_{\mathrm{rx}})^2}. \end{aligned} \tag{3}$$

The free-space term follows the Friis distance dependence, while the coherent correction retains the phase difference between the direct and reflected paths [10]–[12]:

$$\mathrm{FSPL} = 32.45 + 20\log_{10}(d_{\mathrm{LoS}}/1000) + 20\log_{10}(f_{\mathrm{MHz}}), \tag{4}$$

$$C_{2\mathrm{ray}} = -20\log_{10}\left|1 + \rho(h_{\mathrm{tx}})\frac{d_{\mathrm{LoS}}}{d_{\mathrm{ref}}} e^{-j[k(d_{\mathrm{ref}} - d_{\mathrm{LoS}}) + \phi(h_{\mathrm{tx}})]}\right|. \tag{5}$$

Here $k = 2\pi f/c$ is the free-space wavenumber, ρ is an effective reflected-field amplitude, and ϕ is a fitted phase offset. The latter two are CKM calibration terms, not material Fresnel coefficients. The deployed LoS prior is

$$\mathrm{PL}_{\mathrm{LoS}} = \mathrm{FSPL} + C_{2\mathrm{ray}} + \beta_{\mathrm{LoS}}(h_{\mathrm{tx}}) + r(d_{2D}, h_{\mathrm{tx}}). \tag{6}$$

Calibration uses valid LoS pixels from calibration-side cities only. In each 5 m height bin, a small grid search selects effective amplitude and phase. A closed-form bias removes the remaining vertical offset. Finally, a smoothed radial table stores the mean residual left by the coherent model. Neighboring height bins are smoothed and linearly interpolated at inference. This structured sequence targets the constructive/destructive rings visible around the centered transmitter without giving the prior enough flexibility to memorize building layouts.

In implementation terms, the LoS correction is a lookup table. The stored entries are the effective ρ, ϕ, and bias curves indexed by transmitter-height bin, together with r indexed by height and radial-distance bin. Inference linearly interpolates neighboring height entries and reads the radial correction for each receiver; it does not refit on the new map. This distinction matters: although the table is data-calibrated, all 14 reported test cities are unseen during table construction. The free-space/two-ray structure supplies extrapolatable geometry, while the lookup table supplies a compact correction learned on other environments.

For each height bin, candidates $\hat{\rho} \in \{0, 0.05, \ldots, 1.5\}$ and 48 uniformly spaced phases in $[-\pi, \pi)$ are evaluated. For a candidate pair, the remaining bias is

$$\hat{\beta}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_h|} \sum_{i \in B_h} \left[y_i - \mathrm{FSPL}_i - C_{2\mathrm{ray},i}(\hat{\rho}, \hat{\phi}) \right]. \tag{7}$$

The pair minimizing the bin RMSE is retained. After smoothing the height curves, the remaining error is averaged by radius and stored as r. This ordering prevents the table from encoding arbitrary two-dimensional building texture.

B. NLoS Branch

The blocked-link branch starts from a COST231/Hata-derived urban trend [6]. Its source domain does not match this 7.125 GHz CKM, so the expression is used as an inspectable basis function rather than a transferred final law. Each map has 513×513 cells at 1 m spacing, with the transmitter projection at cell $(256, 256)$. For a receiver cell $p = (u, v)$, the implementation first computes

$$\begin{aligned} d_{2D}(p) &= 1\,\mathrm{m}\sqrt{(u-256)^2 + (v-256)^2}, \\ d(p) &= \max\{d_{2D}(p), 1\,\mathrm{m}\}, \\ h_t &= \max\{h_{\mathrm{tx}}, 1\,\mathrm{m}\}, \quad h_r = \max\{h_{\mathrm{rx}}, 0.5\,\mathrm{m}\} = 1.5\,\mathrm{m}. \end{aligned} \tag{8}$$

Thus $d_{\mathrm{km}} = \max\{d(p)/1000\,\mathrm{m}, 0.001\}$; all logarithms below receive the numerical value in the stated unit. With $a(h_{\mathrm{rx}}) = (1.1 \log_{10} f_{\mathrm{MHz}} - 0.7) h_{\mathrm{rx}} - (1.56 \log_{10} f_{\mathrm{MHz}} - 0.8)$, and $\eta(h_t) = 44.9 - 6.55 \log_{10} h_t$, define

$$\begin{aligned} \widetilde{\mathrm{PL}}_C = {} & 46.3 + 33.9 \log_{10} f_{\mathrm{MHz}} - 13.82 \log_{10} h_t - a(h_r) \\ & + \eta(h_t) \log_{10} d_{\mathrm{km}} + 3. \end{aligned} \tag{9}$$

The $+3$ dB metropolitan correction is retained inside this empirical basis; Eq. (9) is not claimed to be a directly valid COST231 deployment law at the carrier frequency and heights considered here. The regression input is the clipped map

$$\mathrm{PL}_C = \mathrm{clip}\left(\widetilde{\mathrm{PL}}_C, 0, 185\right). \tag{10}$$

The clipping is applied before PL_C enters the regression. For the elevation-dependent morphology features below, let $\theta(p) = (180/\pi) \tan^{-1}[(h_t - h_r)/d(p)]$.

The visibility mask identifies that a direct path is blocked but not which street canyon, roof edge, or clutter configuration dominates the remaining path. We therefore augment PL_C with exactly defined local quantities. Let $B(p) = \mathbf{1}\{T(p) \neq 0\}$ be the building indicator from topology height T; let $G = 1 - B$; and let $N(p) = \mathbf{1}\{m_{\mathrm{LoS}}(p) \leq 0.5\} G(p)$ be the NLoS-ground indicator. For odd k, define the reflect-padded box mean

$$\mathcal{A}_k[z](p) = \frac{1}{k^2} \sum_{q \in \mathcal{W}_k(p)} z(q). \tag{11}$$

The implemented features are

$$\begin{aligned} \ell_d &= \ln(1 + d_{2D}/1\,\mathrm{m}), & \delta_k &= \mathcal{A}_k[B], \\ h_k &= \mathcal{A}_k[TB]/90\,\mathrm{m}, & n_k &= \mathcal{A}_k[N], \\ \sigma_{\mathrm{sh}} &= 2.3197[\max(90 - \theta, 0)]^{0.2361}, & & \\ \theta_n &= \mathrm{clip}(\theta/90, 0, 1), & k &\in \{15, 41\}. \end{aligned} \tag{12}$$

The shadowing proxy is an empirical elevation feature; its role here is deterministic feature construction, not stochastic shadow-fading generation. Thus, h_k is the window-averaged building height divided by 90 m, including ground pixels as zeros; it is not the mean conditioned only on buildings. The final vector, in implementation order, is

$$\begin{aligned} \mathbf{x} = [&\mathrm{PL}_C, \ell_d, \delta_{15}, \delta_{41}, h_{15}, h_{41}, \delta_{41}\ell_d, \\ & n_{15}, n_{41}, n_{41}\ell_d, \sigma_{\mathrm{sh}}, \theta_n, n_{41}\theta_n, 1]^\top. \end{aligned} \tag{13}$$

Table II removes any remaining ambiguity by giving the array operation, scale, and interpretation of every regressor. Window sizes are in pixels and therefore metres here; $\mathcal{A}_k$ always includes all k^2 cells after reflect padding, including zeros over ground in the height features. Training standardizes non-intercept features within each fitted regime; coefficients are then transformed back to the raw feature space for the deployed dot product.

The regression regime is observable. From a complete topology map, let D be its building-pixel fraction and H its mean nonzero building height. The class is dense/high-rise if $D \geq 0.2549$ or $H \geq 15.95$ m; open/low-rise if $D \leq 0.1957$ and $H \leq 10.91$ m; and mixed/mid-rise otherwise. Transmitter-height bins are low for $h_{\mathrm{tx}} \leq 58.12$ m, middle up to 103.85 m, and high above it. For each training map, at most 1024 NLoS pixels are selected with a deterministic sample seed (global seed 1234). Let $\mathbf{z}$ contain the standardized non-intercept features and an unpenalized intercept. For regime q, the fitted ridge model is

$$\begin{aligned} \hat{\mathbf{w}}_q &= \arg\min_{\mathbf{w}} \sum_{i \in q} (\mathbf{z}_i^\top \mathbf{w} - y_i)^2 + 10^{-2} \|\mathbf{w}_{\neg b}\|_2^2, \\ \mathrm{PL}_{\mathrm{NLoS}} &= \mathrm{clip}(\mathbf{x}^\top \hat{\mathbf{w}}_q, 20, 185). \end{aligned} \tag{14}$$

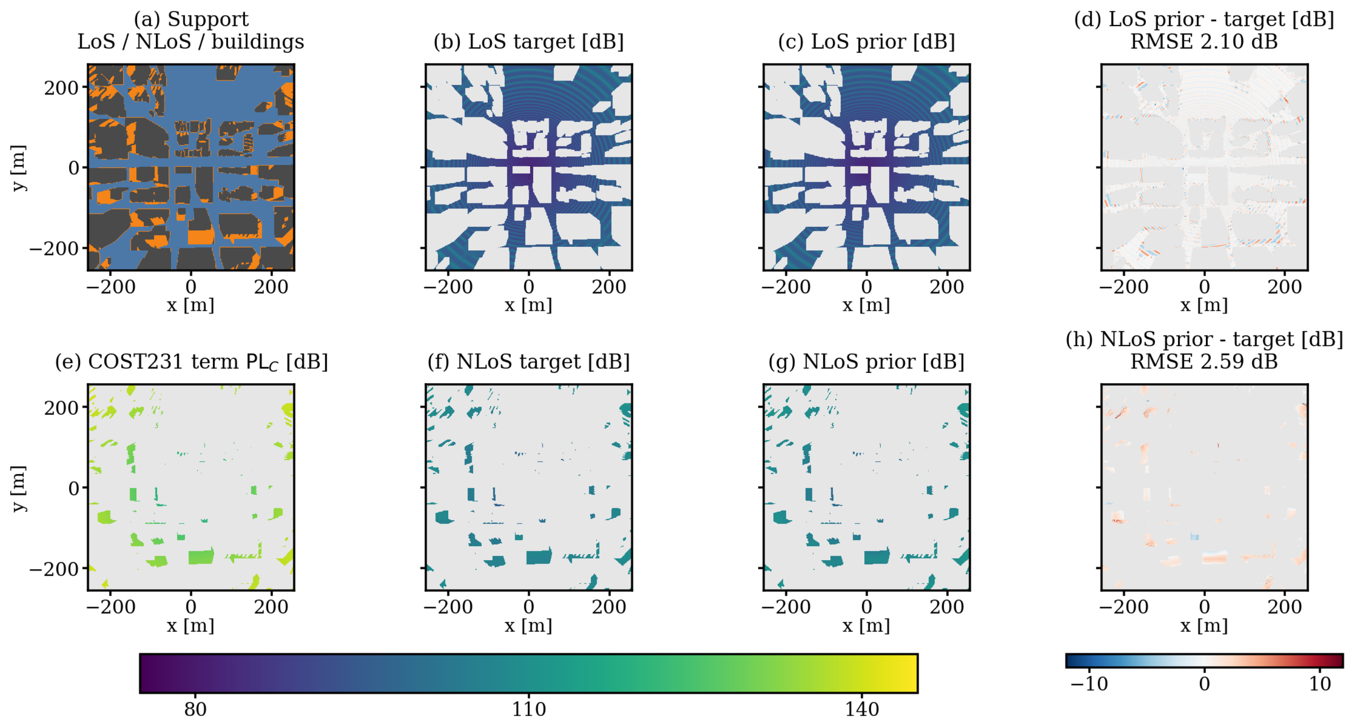

Fig. 2. A held-out example from Vancouver ("sample_15262", $h_{\mathrm{tx}} = 50.30\,\mathrm{m}$, dense/high-rise, low-transmitter regime). Panel (a) shows the geometric support; (b)–(d) are the LoS target, frozen lookup-table prior, and signed error; (e) visualizes the raw COST231 term PL_C; and (f)–(h) give the NLoS target, calibrated prior, and signed error. Grey excludes pixels outside the displayed regime or without a valid target. The unseen map contains 119180 LoS and 25607 NLoS receivers, with 2.10 dB and 2.59 dB RMSE, respectively.

TABLE II
EXACT CONSTRUCTION OF EVERY NLOS MULTILINEAR-REGRESSION INPUT AT RECEIVER CELL p. THE FEATURE ORDER IS ALSO THE COEFFICIENT-VECTOR ORDER. "UNITLESS" MEANS THAT THE STATED METRE OR DEGREE NORMALIZATION HAS ALREADY BEEN APPLIED.

j	Feature $x_j(p)$	Exact computation	Scale and role
1	PL_C	Eq. (10)	dB; clipped COST231 term
2	ℓ_d	$\ln[1 + d_{2D}(p)/1\,\mathrm{m}]$	unitless; radial growth
3	δ_{15}	$15^{-2}\sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; local building fraction
4	δ_{41}	$41^{-2}\sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{T(q) \neq 0\}$	unitless; contextual building fraction
5	h_{15}	$(15^2\,90\,\mathrm{m})^{-1}\sum_{q\in\mathcal{W}_{15}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; local height mass
6	h_{41}	$(41^2\,90\,\mathrm{m})^{-1}\sum_{q\in\mathcal{W}_{41}(p)} T(q)\mathbf{1}\{T(q) \neq 0\}$	unitless; contextual height mass
7	$\delta_{41}\ell_d$	$\delta_{41}(p)\ell_d(p)$	unitless; density–range interaction
8	n_{15}	$15^{-2}\sum_{q\in\mathcal{W}_{15}(p)} \mathbf{1}\{m_{\mathrm{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; local blocked-ground fraction
9	n_{41}	$41^{-2}\sum_{q\in\mathcal{W}_{41}(p)} \mathbf{1}\{m_{\mathrm{LoS}}(q) \leq 0.5\}\mathbf{1}\{T(q) = 0\}$	unitless; contextual blocked-ground fraction
10	$n_{41}\ell_d$	$n_{41}(p)\ell_d(p)$	unitless; blockage–range interaction
11	σ_{sh}	$2.3197[\max\{90 - \theta(p), 0\}]^{0.2361}$	dB; elevation-dependent shadowing proxy
12	θ_n	$\mathrm{clip}[\theta(p)/90, 0, 1]$	unitless; normalized elevation angle
13	$n_{41}\theta_n$	$n_{41}(p)\theta_n(p)$	unitless; blockage–elevation interaction
14	1	an all-ones array with the same 513×513 shape	unitless; intercept

In the second line, $\hat{\mathbf{w}}_q$ denotes the algebraically equivalent raw-space coefficients after undoing standardization. All nine topology and height regimes are fitted on the 10840 training maps only. Table III gives representative coefficients for the middle transmitter-height bin. Their magnitudes must be read together with the feature scales in Table II; they are not feature importance scores. The accompanying CSV contains all nine 14-term vectors at machine precision.

The current artifact contains all 3×3 NLoS topology/height vectors; the inference code checks compatible height-bin fallbacks only if an exact key is absent. This branch is semi-empirical, but its raw estimate, each feature, selected regime, coefficient, and pre-clipping contribution remain auditable.

The complete prior uses the deterministic LoS mask m_{LoS}:

$$\mathrm{PL}_{\mathrm{prior}} = m_{\mathrm{LoS}}\mathrm{PL}_{\mathrm{LoS}} + (1 - m_{\mathrm{LoS}})\mathrm{PL}_{\mathrm{NLoS}}. \tag{15}$$

Both deployed branches are finally clipped to the target support $[20, 185]$ dB.

TABLE III
SELECTED RAW-SPACE NLOS COEFFICIENTS FOR THE MIDDLE TRANSMITTER-HEIGHT BIN.

Feature	Open/low-rise	Mixed/mid-rise	Dense/high-rise
PL_C	-0.117224	0.109671	0.117467
ℓ_d	8.669730	4.265489	3.320933
δ_{41}	-14.501030	-29.259829	-33.524686
n_{41}	15.473961	11.327532	7.311767
σ_{sh}	-10.152455	-7.352513	-11.289259
θ_n	-20.718816	-13.965305	-22.991302
Bias	144.446889	119.161245	150.024085

TABLE IV
FROZEN PRIOR ON THE 2590-MAP, 14-CITY TEST SET. INTERVALS RESAMPLE CITIES, NOT PIXELS.

Metric	Estimate	95% interval
Overall RMSE [dB]	1.9277	[1.7960, 2.0967]
Overall MAE [dB]	1.2151	[1.1350, 1.3174]
LoS RMSE [dB]	1.7370	[1.6603, 1.8225]
NLoS RMSE [dB]	3.5275	[3.2106, 3.8529]
Map correlation	0.948119	[0.942648, 0.952578]
Macro city RMSE [dB]	1.9823	[1.8484, 2.1293]

TABLE V
LARGEST NLOS TERMS BY MEAN ABSOLUTE PRE-CLIPPING CONTRIBUTION OVER ALL 31 365 314 VALID NLOS TEST RECEIVERS. VALUES ARE IN DB.

Feature	Mean contribution	Mean magnitude
Bias	149.774	149.774
σ_{sh}	-72.725	72.725
ℓ_d	20.188	20.188
PL_C	13.826	18.078
δ_{41}	-6.287	6.310
$\delta_{41}\ell_d$	4.948	4.974
θ_n	-4.454	4.454
n_{41}	1.771	3.752

TABLE VI
PRIOR RMSE BY HEIGHT ON THE 14-CITY FINAL TEST SET.

Height	Overall	LoS	NLoS	Maps
12–50 m	2.1876	1.7943	3.8133	624
50–150 m	1.9538	1.7788	3.4256	1385
150–300 m	1.6557	1.6370	2.3584	387
300–500 m	1.5591	1.5496	2.2827	194

IV. RESULTS

The component ablation uses the 2750-map validation partition; every fitted variant still uses training cities only. The coherent term reduces LoS RMSE from 3.7900 to 1.7412 dB; the radial table gives a smaller further reduction. The raw COST231 term is not accurate enough by itself at 25.4044 dB. A global NLoS ridge model reduces this to 3.2179 dB, while observable topology and height regimes reach 3.1879 dB.

Table IV gives the final attenuation-only test result. The prior reaches 1.9277 dB RMSE and 1.2151 dB MAE across all valid receiver pixels. The map correlation of 0.948119 shows that its accuracy is not obtained only by matching a global offset: relative high- and low-attenuation regions are generally placed correctly. The LoS error is 1.7370 dB. NLoS remains the hard regime at 3.5275 dB, consistent with the unobserved identity of the dominant diffracted or reflected path after direct-path blockage.

Per-city RMSE ranges from 1.6097 dB in Halifax to 2.5827 dB in Osaka; the unweighted 14-city mean is 1.9823 dB. This spread and the city-bootstrap interval prevent the global score from being read as uniform performance across environments.

After excluding buildings and ground pixels without a valid target, the test metric covers 423 087 119 receivers, of which 7.413% are NLoS. Although the pixel-weighted score is therefore LoS dominated, 424 of the 2590 test maps contain more than 25% NLoS receivers, motivating the separate regime metrics.

Table V complements the raw coefficients with their actual test-set scale. For each NLoS receiver, the selected topology and transmitter-height regime supplies one coefficient for each feature. Multiplying that coefficient by the receiver's feature value gives its applied term before clipping. "Mean contribution" averages the signed applied terms; "mean magnitude" averages their absolute values. Positive and negative terms therefore cancel only in the first column. Every NLoS receiver has equal weight, and all displayed features are nonnegative, so a gap between the columns means that the fitted coefficient changes sign among regimes.

The sign changes are not determined by topology alone. The PL_C coefficient is negative for dense/high-rise at high transmitter height, mixed/mid-rise at high height, and open/low-rise at middle height, but positive in the other six regimes. Across the complete NLoS test set, positive PL_C terms add 15.952 dB to the pixel-weighted average and negative terms subtract 2.126 dB. The signed result is therefore $15.952 - 2.126 = 13.826\,\mathrm{dB}$, while ignoring the signs gives $15.952 + 2.126 = 18.078\,\mathrm{dB}$. The n_{41} coefficient is negative at low transmitter height in all three topologies and for open/low-rise at high height, but positive in the other five regimes. δ_{41} and $\delta_{41}\ell_d$ reverse sign only for open/low-rise at high transmitter height, so their cancellation is small. Bias, σ_{sh}, ℓ_d, and θ_n retain one sign in all nine regimes. Marginal averages of coefficients and features are therefore omitted: their pairing changes across regimes. These remain scale diagnostics, not causal ablations, because correlated regressors can exchange weight.

Height is part of the estimator rather than a post-hoc reporting variable. Table VI shows the frozen prior on the final test cities only for four physical height intervals. Error decreases from 2.1876 dB at 12 m to 50 m to 1.5591 dB above 300 m. The largest change is in NLoS, where low transmitters interact more strongly with local rooftops and street canyons. The trend does not imply that altitude alone solves dense prediction; it shows why a single height-independent calibration would combine physically different regimes.

The result is best interpreted as a strong physical/statistical anchor rather than a universal propagation model. The coefficients are tied to the fixed frequency, antenna assumptions, simulator, and available visibility mask. The city split tests transfer across urban layouts, but the labels still come from ray tracing rather than field measurements.

A. Limitations, Visibility, and Runtime

The visibility mask is the configured MATLAB ray tracer's own binary `LineOfSight` output. Identical city geometry, transmitter position and height, receiver grid, and propagation settings therefore define one mask that can be stored and reused. In a repeated five-condition CPU audit, all 742 392 receiver visibility values were identical between runs.

With this stored ray-traced mask, one final prior map takes a median 0.0602 s (95th percentile 0.0844 s) over 100 maps on an AMD Ryzen 5 5600X CPU; HDF5 reading is timed separately at 0.0046 s median. On the same CPU, with GPU execution disabled for both paths, MATLAB R2024b ray tracing takes a median 102.289 s (range 91.452 s to 260.310 s) over five Barcelona layouts. Each timed run uses one transmitter height, a 1 m receiver grid, one reflection, and no diffraction, and extracts attenuation, delay spread, and angular spread. The raw median wall-clock difference is therefore about 1700$\times$ for the stored-mask prior core. This is not a like-for-like speedup because MATLAB returns three channel quantities, whereas this paper's prior returns only attenuation. Nevertheless, the comparison shows a substantial computational gap at the same spatial resolution. A future journal paper will apply analogous interpretable priors to delay and angular spread, enabling a closer multi-output comparison.

All current scenes use flat terrain. Consequently, the reported visibility, distance, and height relationships do not account for terrain-induced blockage or for changes in local ground elevation. Slopes and ridges can alter both the LoS boundary and the meaning of transmitter height relative to nearby receivers. Follow-up work will introduce non-flat digital terrain, compute antenna heights and reflected-path geometry relative to local elevation, add terrain descriptors to the NLoS calibration, and evaluate transfer across relief classes.

V. Conclusion

This paper presented an attenuation prior for dense, variable-height UAV CKMs. Rather than treating attenuation as generic image-to-image regression, it assigns stable structure to explicit components: coherent radial and two-ray behaviour in LoS, COST231 propagation and local morphology in NLoS, and a deterministic visibility mask to route between them. Every lookup table and regression coefficient is fitted on the training cities and frozen before the unseen cities are evaluated, so the final test result requires no city-specific adaptation.

On 2590 maps from 14 held-out cities, the prior reaches 1.9277 dB overall RMSE, 1.7370 dB LoS RMSE, 3.5275 dB NLoS RMSE, and 0.948119 map correlation. The component ablation shows that the coherent LoS structure and calibrated NLoS morphology are complementary rather than interchangeable. Separate regime reporting is essential: the aggregate metric is LoS dominated, while hundreds of maps contain substantial NLoS support.

The broader conclusion is that a strong CKM estimator should explain stable physical and statistical structure before a more flexible model is asked to learn the remaining local detail. Here that principle yields a compact, auditable CPU estimator with a median runtime of 0.0602 s per map. The prior can therefore serve both as a standalone surrogate and as a frozen anchor for residual learning. Future work will incorporate non-flat terrain, study transfer across frequencies and antenna configurations, and extend the same prior-first methodology to delay and angular spread in a future journal paper.

References

[1] Y. Zeng and X. Xu, "Toward environment-aware 6G communications via channel knowledge map," *IEEE Wireless Communications*, vol. 28, no. 3, pp. 84–91, 2021.

[2] Y. Zeng, J. Chen, J. Xu, D. Wu, X. Xu, S. Jin, X. Gao, D. Gesbert, S. Cui, and R. Zhang, "A tutorial on environment-aware communications via channel knowledge map for 6G," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 3, pp. 1478–1519, 2024.

[3] W. Khawaja, I. Guvenc, D. W. Matolak, U.-C. Fiebig, and N. Schneckenberger, "A survey of air-to-ground propagation channel modeling for unmanned aerial vehicles," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2361–2391, 2019.

[4] A. Saboor and E. Vinogradov, "Millimeter-wave UAV channel model with height-dependent path loss and shadowing in urban scenarios," arXiv:2511.10763, 2025. [Online]. Available: https://arxiv.org/abs/2511.10763

[5] J.-H. Lee, O. G. Serbetci, D. P. Selvam, and A. F. Molisch, "PMNet: Robust pathloss map prediction via supervised learning," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2023. [Online]. Available: https://arxiv.org/abs/2211.10527

[6] COST Action 231, "Digital mobile radio towards future generation systems: Final report," Office for Official Publications of the European Communities, Luxembourg, Tech. Rep., 1999. [Online]. Available: https://op.europa.eu/en/publication-detail/-/publication/f2f42003-4028-4496-af95-beaa38fd475f

[7] R. Levie, Çağkan Yapar, G. Kutyniok, and G. Caire, "RadioUNet: Fast radio map estimation with convolutional neural networks," *IEEE Transactions on Wireless Communications*, vol. 20, no. 6, pp. 4001–4015, 2021. [Online]. Available: https://arxiv.org/abs/1911.09002

[8] J.-H. Lee, J. Lee, S.-H. Lee, and A. F. Molisch, "PMNet: Large-scale channel prediction system for ICASSP 2023 first pathloss radio map prediction challenge," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–2. [Online]. Available: https://doi.org/10.1109/ICASSP49357.2023.10095257

[9] Çağkan Yapar, F. Jaensch, R. Levie, G. Kutyniok, and G. Caire, "The first pathloss radio map prediction challenge," in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–2. [Online]. Available: https://doi.org/10.1109/ICASSP49357.2023.10433928

[10] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Prentice Hall, 2002. [Online]. Available: https://www.pearson.com/en-us/subject-catalog/p/wireless-communications-principles-and-practice/P200000003199

[11] A. Goldsmith, *Wireless Communications*. Cambridge University Press, 2005.

[12] C.-C. Chiu, A.-H. Tsai, H.-P. Lin, C.-Y. Lee, and L.-C. Wang, "Channel modeling of air-to-ground signal measurement with two-ray ground-reflection model for UAV communication systems," in *2021 30th Wireless and Optical Communications Conference (WOCC)*, 2021, pp. 251–256.